VANI persona system: current architecture

Seller GLID → context folder → Claude builds persona → Sarvam voice agent calls → results feed learning loops. As of commit a8f2e06 (2026-10-10). Real seller data stays in git-ignored `data/`.

Claude LLM call · Sarvam service / Sarvam LLM · Script / file (ours) · **Red = PII boundary: pseudonymised before any frontier-model call**

1 · DATA SUPPLIED (SOURCES)

Offline files (data/, git-ignored)
PS07 persona set: seller file (149k), bot calls (269k), evaluator outputs, call turns, executive calls
Global Context (5k sellers): enquiries, buy leads bought, WhatsApp, exec calls, profile
Best-Time-to-Call (531k attempts)
Seller Onboarding: locations, categories, products, KYC, catalogue health

Internal APIs (prod hosts)
imdwh: category value (mcatpriceapi, mcatetoretail)
users SOA: users_detail, pnssetting, filtercount, profile sections (AK + token via env)
company API: showroom page (C09)
scorecards, query/enquiry summaries

Redash → pg-imblr (read-only SQL)
Buyer AOV (eto_attribute)
Peer-stat ladder (E11)
Showcase buy leads, live + expired (E12)
City tier (gl_city)
GST state code (X06, new, first 2 digits only)

Team decisions & priors (raw/)
Identity line (never "AI"), money-lead hook, peer-comparison rule, gender/voice priors, hook_priors from PS07 bot-version experiment, 21 role-play call analyses, call-flow playbook (12 objections)

▼

2 · CONTEXT BUILDER (LAPTOP, VPN) — NO LLM

seller_context_builder.py
+ app/context/SCRUBBER.py, extraction_map.py, http_util.py
index → sellers.sqlite; build → folder; `--as-of` for backtests; `--no-api` offline

→

data/contexts/<glid>/
INDEX.md (key facts K1…, call table, file map)
core.md, business.md, catalogue.md, buyleads.md, enquiries.md, channels.md, peers.md, candidate_leads.md
calls/C01… (turn-numbered), context.json

→

export_for_agent.py + pii_check.py
Hard PII gate: names, competitors → tokens (HMAC pseudonym folder); phones, emails, IDs, pins dropped; export refuses on any hit
→ data/agent_export/<pseudo>/ + name map (stays on premises)

▼

3 · PERSONA DESIGN (RUN_PERSONA.PY, ONE SELLER PER RUN; LAPTOP NOW, DAYTONA SANDBOX LATER)

facts.py
(script)
traits, states, guards, hook eligibility, each with evidence ids
→ facts.json

→

prior.py
(script)
Scoreboard A: persona P0–P6 (all → division → band, +zone, +AOV; κ=20)
Scoreboard B: voice F/M
Thompson draw, 15% P0 holdout → prior.json
data/scoreboards/ (empty today)

→

Claude Opus 5.5 — `claude -p` + skill vani-persona
Reads INDEX, core, facts, prior; opens only needed files
Guards first → division reference → review traits/states → decide hook (primary/backup, cited) → pick top-2 persona → amend from fixed menu → voice, language, words
Writes 02_review.json, 04_persona.json, status.json
Gates (hooks): pre_gate (protects files), log_read (read log), stop_gate (runs validator)

→

validate_persona.py
(gate)
cites exist + quote ≤20 words; banned words (AI, bot, branch manager…); gender agreement; numbers only from verified_facts; money-lead rule; confidence ≤ completeness

→

compile_vars.py + render_prompt.py
Persona → Sarvam variables; checks voice id, pace, pitch, language vs allowed list; names filled back on-prem
→ 05_sarvam_vars.json, 06_sarvam_prompt.md

Skill files supplied to the agent
SKILL.md (steps), references/division_nsd.md, division_csd.md (hooks, 12/11 objection replies, Sarvam blocks), persona_library.json (P0–P6, seed leans), voice_and_language.md, evidence/hook_priors.json, formats/{persona,hook,trait,state}.json, sandbox/CLAUDE.md

Planned / design docs
persona-e2e-plan.md, persona-skill-design.md, VANI persona architecture v6, traits-states library draft (stale: facts.py wins), seller-context final plan, outputs/objections.json, persona-hierarchy docs

▼ variables + prompt sections

4 · SARVAM (VOICE ONLY: SPEECH, TTS, CALL RUNTIME) — AGENT VANI-PERSON V2 (PUBLISHED)

Agent config (files → authoring API)
app/sarvam/vani_prompt.md: ONE fixed prompt, 7 sections filled by `{{prompt_*}}` variables
agent.json settings: patient endpointing, language switch on
app/contracts/sarvam_vars.contract.json (all variables)
Created/updated via console authoring API (Bearer login). **SDK cannot create/publish.**
MCP (mcp.sarvam.ai) possible, not signed in.

→

Call runtime, each turn
Seller speech → **STT** (Saaras) → **Sarvam LLM** (agent LLM + hidden tool-LLM) → may call a tool → **Bulbul v4 TTS** (voice id, pace, pitch from OnStart) → seller
Test route: SDK voice session (voice_smoke.py, text/audio); phone/telephony later

→

Code tools (app/sarvam/vani_tools.py)
OnStart (call start, no LLM): voice, pace, pitch, start language, opening line
read_seller_signal (LLM-triggered every turn): lex-v2 keyword classifier → seller_mode + directive (rushed, frustrated, confused, interested, bot_question, opt_out, language_switch) + switch_log
book_meeting: meeting_fixed + slot

→

After the call (Sarvam analytics)
Post-call labels: call_outcome, main_objection, hook_reaction, persona_fit, seller_mood_end, end_language
Transcript, agent variables, audio URL, duration, ended_by
Built-in language detection/switch stays on mid-call

▼ interaction id

5 · LEARNING LOOPS (BETWEEN CALLS, NEVER DURING A CALL)

collect_call.py
Analytics API (X-API-Key) → 06_call.json, 07_outcome.json

→

review_call.py — Claude Opus 5.5
Frontier judge → 08_review.json: per-turn true mode vs tool mode, persona-fit issues by dimension, hook landed, QA **Fatal** (banned word, unverified number, claimed human, pushed after opt-out, named buyer) / non-fatal

→

Loop 2: loop2_update.py
Adds outcome to scoreboards A and B (all, division, band, zone, AOV, voice keys). Test calls → data/scoreboards_test

→

Loop 1: loop1_propose.py
Finds missed/wrong switches → proposes keyword/directive changes → gold-set check (data/gold, 213 turns; lex-v2 = 0.60) → human approves → new tools version uploaded

→

close_loop.py
Runs the chain end to end on one interaction

HOW WE USE SARVAM (SUMMARY)

- **Voice agent runtime:** STT + Sarvam LLM + Bulbul v4 TTS, hosted by Sarvam.
- **Persona reaches Sarvam only as variables** (one fixed prompt; no per-seller prompt edits).
- **Start of call:** OnStart applies voice, pace, pitch, language, opening.
- **During call:** LLM calls read_seller_signal each turn; directive overrides style for that reply only; Sarvam auto language switch handles language.
- **After call:** Sarvam labels + analytics API; Claude reviews.
- **Not used for:** persona design or review (Claude does those). Sarvam Chat 105B not used for decisions.

ALL LLM CALLS

- **Claude Opus 5.5** (`claude -p`): persona design per seller (~6 min); post-call review.
- **Sarvam LLM** (in call): conversation + tool choice; not configurable (no BYO LLM).
- **Claude Sonnet 5.5** (this session): analysis, docs, fact checks (offline).
- **Jev** (LiteLLM gateway): researched as a mid-call classifier; cut from the demo.
- No LLM in: builder, facts.py, prior.py, validator, compile_vars, loop2.

BOUNDARIES & RULES

- Secrets (AK, tokens, Sarvam keys) only in env / git-ignored .env; never stored in docs.
- Agent sees only the pseudonymised folder; names filled on premises after validation.
- Never say AI/bot/branch manager; sidestep once, "virtual assistant" if pressed; never claim human.
- Numbers only from verified_facts; GSTIN/PAN/phones never in the folder.
- Live API/DB values are not as-of safe for backtests.
- Persona library v0: approval pending.